



K. J. Somaiya College of Engineering, Mumbai-77

Batch: A-4 **Roll No.:** 16010122151

Experiment / assignment / tutorial No:- 2

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

TITLE: Software Architecture of any operating system, using ADL

AIM: Study of architecture of operating system .

Expected OUTCOME of Experiment:

Using ADL represent operating system architecture

CO 1. Design the architecture of software systems in various architectural styles.

Books/ Journals/ Websites referred:

- <https://www.xcubelabs.com/blog/software-architecture-understanding-styles-and-patterns-for-effective-system-design/#:~:text=Architectural>
- https://www.tutorialspoint.com/operating_system/os_architecture.htm
- <https://www.geeksforgeeks.org/android/android-architecture/>

Background Theory:

A software system's high-level structure, or software architecture, is made up of the rules, patterns, and guidelines that determine how the system is organized, interacts, and is related to its components.

The design decisions will help in providing:

Modularity: allowing the system into interchangeable components that can be developed, tested, and maintained independently.

Encapsulation: Hiding the details of the components, exposing only the necessary information, thus reducing the complexity of the system.

Security: Incorporating the measures to protect the system against the unauthorized access.



Documentation: Provides clear documentation of the system architecture, thus facilitating communication and better understanding of the system.



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Performance: Ensuring that the system meets the required performance metrics such as resource utilization, throughput, etc.

There are various types of Software Architectural Styles.

- Traditional, language-influenced styles
 - Main program and subroutines
 - Object-oriented
- Layered
 - Virtual machines
 - Client-server
- Data-flow styles
 - Batch sequential
 - Pipe and filter
- Shared memory
 - Blackboard
 - Rule based
- Interpreter
 - Interpreter
 - Mobile code
- Implicit invocation
 - Event-based
 - Publish-subscribe
- Peer-to-peer
- “Derived” styles
 - C2
 - CORBA
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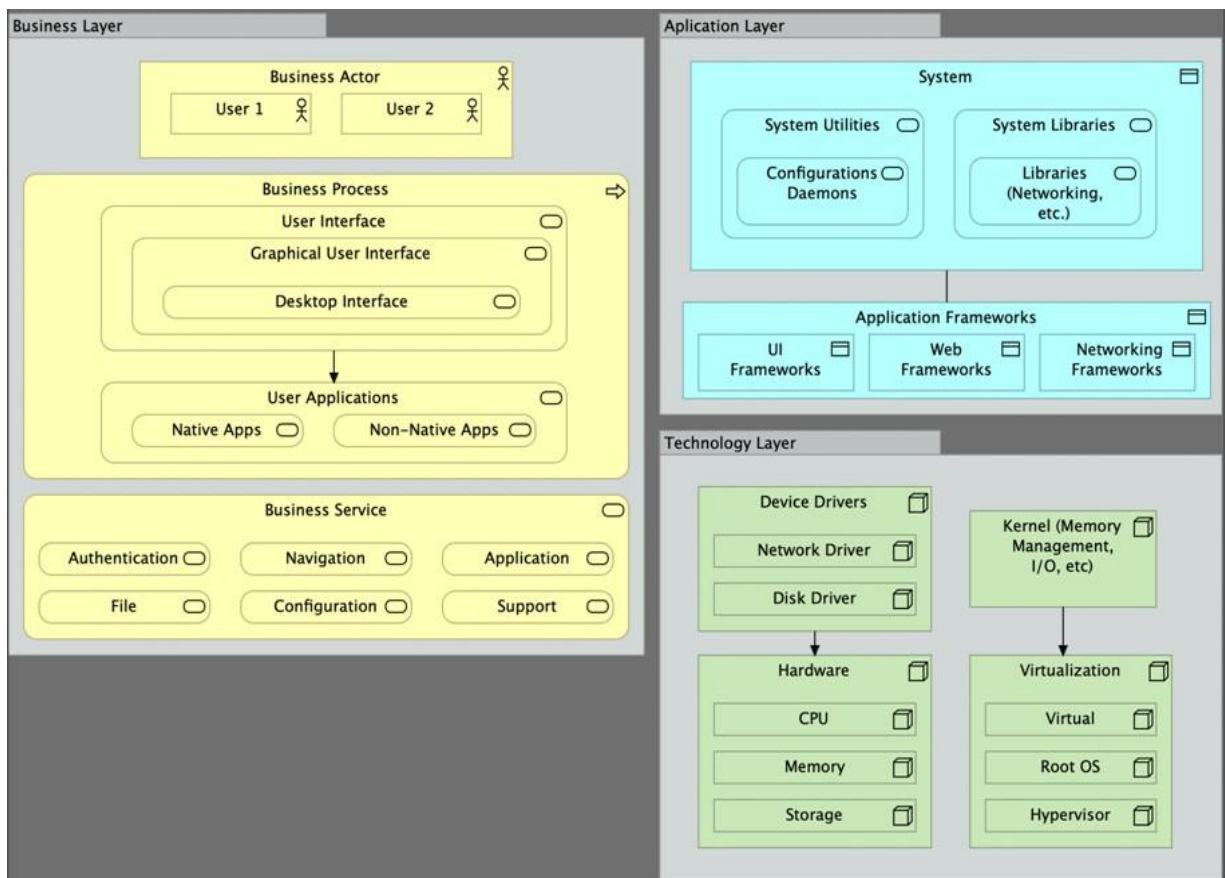
Different architectural patterns will be used for implementing software systems based on: Software architecture patterns offer reusable designs for various situations, to offer advantages such as improved efficiency, productivity, speed, cost optimization, and better planning.

Laboratory Work:

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The previous experiment introduces ArchiMate a tool used for drawing Software architecture design. The students will be required to study architecture of any of the Operating System from the following list and draw diagram

- Ubuntu
- Windows
- macOS



Business Layer

- **Business Actors**
 - *End User Type 1:* One category of individuals who use the system and engage with its features.
 - *End User Type 2:* A distinct class of users who interact with the solution differently.
- **Core Business Processes**



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- *User Interaction Interface*: Mechanisms enabling end users to communicate and work with the system.
- *Visual Interface—Desktop*: The graphical component designed specifically for desktop platforms.
- *User-Focused Applications*: Programs designed to serve user needs.
 - *Platform-Specific (Native) Applications*: Software tailored for seamless operation on Windows environments.
 - *Cross-Platform (Non-Native) Applications*: Programs originally built for other platforms but adapted for Windows compatibility.
- **Business-Oriented Services**
 - *Identity Verification*: Functions dedicated to confirming and managing user identities.
 - *Application Navigation*: Services that direct users through workflows and features.
 - *General App Services*: Foundational support for application operation.
 - *File Operations*: Capabilities related to the management and handling of files.
 - *Settings Management*: Tools for adjusting system or application options.
 - *User Assistance*: Support services aimed at aiding users in resolving issues.

Application Layer

- **System Components**
 - *System Management Utilities*: Tools that streamline and oversee system maintenance tasks.
 - *System Libraries*: Pre-built code modules, including:
 - *Configuration Daemons*: Background processes that handle setup and environmental management.



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- *Functional Libraries (Networking, etc.):* Modules that enable specialized tasks, such as network communication.

- **Application Frameworks**

- *User Interface Frameworks:* Toolkits for crafting interactive front-ends.
- *Web Application Frameworks:* Platforms for building web-based solutions.
- *Networking Frameworks:* Infrastructure facilitating network operations.

Technology Layer

- **Device Interaction**

- *Network Adapters:* Software interfaces for network hardware.
- *Storage Controllers:* Drivers that manage data storage devices.

- **Operating System Core**

- *Kernel:* The foundational part of the OS responsible for memory allocation, input/output control, and system management.

- **Resource Virtualization**

- *Virtual Resource Manager:* Components that abstract and organize virtual resources.
- *Primary Operating System:* The base OS in a virtual environment.
- *Hypervisor:* Orchestrates creation and management of virtual machines.

- **Hardware Elements**

- *Processing Unit:* The CPU handles core computations.
- *Memory Modules:* RAM and related storage for active data.
- *Persistent Storage:* Devices providing long-term data retention, such as hard drives and SSDs.



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Post Laboratory questions:

1. Compare Software Architectural patterns and Design Pattern

Software Architectural Patterns are high-level structural frameworks that define the fundamental organization of an entire software system. They outline major system components, their responsibilities, and how they interact. Architectural patterns have a broad scope—guiding the design, scalability, maintainability, and deployment of complex software systems.

- Examples: Layered architecture, client-server, microservices, event-driven architecture.

Design Patterns, on the other hand, are proven solutions to common design problems or challenges that occur repeatedly at the level of individual modules, components, or classes. They focus on best practices for organizing code within components, rather than for the whole application's structure.

- Examples: Singleton, Observer, Factory, Strategy.

Key Differences:

- *Scope*: Architectural patterns cover system-wide concerns, while design patterns address localized implementation issues.
- *Purpose*: Architectural patterns solve problems related to system structure and behavior; design patterns tackle code-level problems.
- *Impact*: Architectures affect scalability, performance, and maintainability at the system level; design patterns improve code clarity, flexibility, and reuse within smaller units.

2. Explain Various type of Software Architecture Patterns giving example of each

Layered (Tiered) Architecture

- Description: Divides the application into distinct horizontal layers, each with specific roles (e.g., presentation, business logic, data access).
- Example: Classic enterprise web applications split into UI, business logic, and database tiers.

Client-Server Architecture

- Description: The system is split into clients (requesters of services/resources) and servers (providers of services/resources).
- Example: Email applications, where the client fetches and manages messages from the mail server.

Microservices Architecture



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- Description: The application is composed of multiple small, independent services that communicate via APIs.
- Example: E-commerce platforms with separate microservices for orders, payments, inventory, and user profiles.

Event-Driven Architecture

- Description: Components communicate by producing and consuming events, typically using a message broker or event bus.
- Example: Online ticket booking systems, where payment completion triggers multiple downstream actions.

Model-View-Controller (MVC)

- Description: Separates an application into three main logical components—Model (data), View (User Interface), and Controller (business logic).
- Example: Web applications using frameworks like Ruby on Rails or ASP.NET MVC.

Service-Oriented Architecture (SOA)

- Description: Arranges software as a set of reusable services, accessible over a network, to promote reusability and interoperability.
- Example: Large enterprises performing data exchange across departments using SOAP web services.

Peer-to-Peer Architecture

- Description: All nodes (peers) act as both clients and servers, sharing resources directly without reliance on a centralized server.
- Example: File sharing applications like BitTorrent.

3. Explain in brief the architecture of Android Operating system.

The Android OS is structured in layers, each fulfilling specific system responsibilities:

Linux Kernel Layer

- Role: Acts as the foundational layer, providing core system services, device drivers (for camera, audio, etc.), power management, and security features.

Hardware Abstraction Layer (HAL)



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- Role: Serves as an interface between the kernel and device hardware, standardizing hardware access for higher layers.

Native Libraries and Android Runtime

- Native Libraries: Include C/C++ libraries for essential functionalities, such as SQLite (database), WebKit (browser engine), OpenGL ES (graphics).
- Android Runtime: Runs apps using the ART (Android Runtime) or, historically, the Dalvik VM. Provides core libraries for running applications written in Java or Kotlin.

Application Framework

- Role: Offers reusable Java classes and services (e.g., Activity Manager, Notification Manager, Resource Manager) that developers use to build apps.

Application Layer

- Role: Contains end-user applications—apps that run in user space (phone dialer, SMS, browser, third-party apps).